

Bai–Perron Theory Workbook: Conditional-Design Rebuild

Profile Sieve Empirical Likelihood with Nuisance Breaks

One-break estimated-partition Wilks, unknown-order selection, and proof map

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1 Primitive Setup and Notation

For a scalar array $\mathbf{a} = (a_1, \dots, a_T)'$, write

$$\|\mathbf{a}\|_T^2 = T^{-1} \sum_{t=1}^T a_t^2, \quad \|\mathbf{a}\|_\infty = \max_{t \leq T} |a_t|.$$

All $O_p(\cdot)$ and $o_p(\cdot)$ statements for vectors and matrices use Euclidean and spectral norm unless another norm is displayed.

The maintained null model is

$$Y_t = \beta_0 X_{t-1} + \sum_{j=0}^{q_0} m_j(W_{t-1}) \mathbf{1}\{k_{j,0} < t \leq k_{j+1,0}\} + U_t, \quad (1.1)$$

where

$$0 = k_{0,0} < k_{1,0} < \dots < k_{q_0,0} < k_{q_0+1,0} = T.$$

Throughout the known-partition theory, q_0 is fixed. The main estimated-partition theorem later specializes to one unknown nuisance break, $q_0 = 1$. Candidate searches use a trimming constant ϵ , while the true partition has slack $\epsilon_0 > \epsilon$:

$$\min_{0 \leq j \leq q_0} (k_{j+1,0} - k_{j,0}) \geq \epsilon_0 T, \quad (\bar{q} + 1)\epsilon < 1.$$

The coefficient β_0 is stable. All structural breaks are in the nuisance component.

For a candidate partition $\Lambda = (k_1, \dots, k_q)$, let $k_0 = 0$, $k_{q+1} = T$, and

$$I_j(\Lambda) = \{t : k_j < t \leq k_{j+1}\}, \quad j_\Lambda(t) = j \text{ if } t \in I_j(\Lambda).$$

For the search trimming constant, define the admissible candidate set

$$\mathcal{L}_q(\epsilon) = \left\{ \Lambda = (k_1, \dots, k_q) : 0 = k_0 < k_1 < \dots < k_q < k_{q+1} = T, \min_{0 \leq j \leq q} (k_{j+1} - k_j) \geq \epsilon T \right\}.$$

Let

$$\mu_t = m_{j_{\Lambda_0}(t)}(W_{t-1}), \quad \boldsymbol{\mu} = (\mu_1, \dots, \mu_T)'. \quad (1.2)$$

Let $\mathbf{P}$ be the $T \times K$ sieve matrix with t -th row

$$\mathbf{P}_K(W_{t-1})' = (p_1(W_{t-1}), \dots, p_K(W_{t-1})).$$

For a partition Λ , let $\mathbf{D}_j(\Lambda)$ be the $T \times T$ diagonal selector matrix with t -th diagonal entry $\mathbf{1}\{t \in I_j(\Lambda)\}$, $j = 0, \dots, q$. Define

$$\mathbf{Q}_{K,\Lambda} = [\mathbf{D}_0(\Lambda)\mathbf{P}, \dots, \mathbf{D}_q(\Lambda)\mathbf{P}], \quad \mathbf{M}_{K,\Lambda} = \mathbf{I}_T - \mathbf{Q}_{K,\Lambda}(\mathbf{Q}_{K,\Lambda}'\mathbf{Q}_{K,\Lambda})^\dagger \mathbf{Q}_{K,\Lambda}'. \quad (1.2)$$

The Moore–Penrose inverse makes $\mathbf{M}_{K,\Lambda}$ algebraically defined even when $\mathbf{Q}_{K,\Lambda}'\mathbf{Q}_{K,\Lambda}$ is singular. Such singularity can arise if a candidate segment is too short relative to K or if the selected basis columns are collinear on that segment; the projection remains defined, but the

asymptotic regularity used below may fail. The theory therefore relies on minimum-spaced admissible partitions and Gram-stability/full-rank conditions on the relevant candidate sets.

For each (β, Λ) , define

$$\hat{\mathbf{u}}(\beta, \Lambda) = \mathbf{M}_{K, \Lambda}(\mathbf{Y} - \beta \mathbf{X}_{lag}), \quad (1.3)$$

$$\mathbf{w}^c(\Lambda) = \mathbf{M}_{K, \Lambda} \mathbf{w}, \quad (1.4)$$

$$\hat{Z}_t(\beta, \Lambda) = \hat{u}_t(\beta, \Lambda) w_t^c(\Lambda). \quad (1.5)$$

Let

$$S_T(\Lambda) = T^{-1/2} \sum_{t=1}^T \hat{Z}_t(\beta_0, \Lambda), \quad V_T(\Lambda) = T^{-1} \sum_{t=1}^T \hat{Z}_t(\beta_0, \Lambda)^2.$$

For a scalar score array $Z_1, \dots, Z_T$ satisfying the convex-hull condition, let λ_T be the unique solution of

$$\sum_{t=1}^T \frac{Z_t}{1 + \lambda_T Z_t} = 0, \quad 1 + \lambda_T Z_t > 0 \quad (t \leq T),$$

and define the twice negative empirical-likelihood log ratio by

$$\ell_T(Z_1, \dots, Z_T) = 2 \sum_{t=1}^T \log(1 + \lambda_T Z_t). \quad (1.6)$$

In particular,

$$\ell_T(\beta, \Lambda) = \ell_T\{\hat{Z}_1(\beta, \Lambda), \dots, \hat{Z}_T(\beta, \Lambda)\}.$$

For an interval $[s, e]$, define the null-imposed segment cost

$$\mathcal{C}_K(s, e; \beta) = \min_{\mathbf{c} \in \mathbb{R}^K} \sum_{t=s}^e \{Y_t - \beta X_{t-1} - \mathbf{P}_K(W_{t-1})' \mathbf{c}\}^2.$$

For $\Lambda = (k_1, \dots, k_q)$, define

$$RSS_K(\beta, \Lambda) = \sum_{j=0}^q \mathcal{C}_K(k_j + 1, k_{j+1}; \beta).$$

All minimizers are measurable selections with a fixed deterministic tie-breaking rule. For known order,

$$\hat{\Lambda}_q(\beta) = \arg \min_{\Lambda \in \mathcal{L}_q(\epsilon)} RSS_K(\beta, \Lambda), \quad \tilde{\Lambda} = \hat{\Lambda}_{q_0}(\beta_0).$$

For unknown order,

$$\hat{q}(\beta) = \arg \min_{0 \leq q \leq \bar{q}} \{RSS_K(\beta, \hat{\Lambda}_q(\beta)) + \text{pen}_T(q, K)\}, \quad \hat{\Lambda}(\beta) = \hat{\Lambda}_{\hat{q}(\beta)}(\beta).$$

Although the Moore–Penrose inverse makes $\mathbf{M}_{K, \Lambda}$ algebraically defined for arbitrary Λ , all theoretical searches are restricted to minimum-spaced admissible partitions $\mathcal{L}_q(\epsilon)$; thus no candidate segment is allowed to be asymptotically shorter than the sieve dimension K , and the Gram matrices used in the proofs are nonsingular on the relevant candidate set.

The sieve dimension is allowed to depend on the sample size and is written $K = K_T$. Throughout, $K_T \rightarrow \infty$, subject to the rate restrictions in Assumptions 3–6 and the local Gram regularity conditions below. The basis is normalized so that its first component is constant, say $p_1(w) \equiv 1$, allowing regime-specific intercept and level shifts to be represented by the block sieve.

For local-power calculations, distinguish the null value used in the statistic from the data-generating slope. The local alternative used in the rebuilt theorem is

$$\beta_T = \beta_0 + \frac{h}{\sqrt{T}}, \quad h \in \mathbb{R},$$

with data generated under P_{β_T} while the profile empirical likelihood is evaluated at β_0 . Equivalently,

$$Y_t - \beta_0 X_{t-1} = \mu_t + U_t + \frac{h}{\sqrt{T}} X_{t-1},$$

with the same nuisance break partition Λ_0 . Section 8 records the corresponding drift conditions.

2 Primitive Assumptions

Assumption 1 (Conditional-on-design stochastic environment). *Let*

$$\mathcal{D}_T = \sigma\{X_{t-1}, W_{t-1}, w_t : 1 \leq t \leq T\}.$$

The inclusion of w_t is deliberate: the residualized empirical-likelihood weight $\mathbf{M}_{K,\Lambda}\mathbf{w}$ must be fixed in the conditional-design arguments. Conditional on $\mathcal{D}_T$, the variables $U_{1,T}, \dots, U_{T,T}$ are independent and satisfy

$$\mathbb{E}(U_{t,T} \mid \mathcal{D}_T) = 0, \quad 0 < \underline{\sigma}^2 \leq \sigma_{t,T}^2 := \mathbb{E}(U_{t,T}^2 \mid \mathcal{D}_T) \leq \bar{\sigma}^2 < \infty.$$

Moreover, for every $s \in \mathbb{R}$,

$$\mathbb{E}\{\exp(sU_{t,T}) \mid \mathcal{D}_T\} \leq \exp(Cs^2/2)$$

almost surely, uniformly in t and T . The subscript T is suppressed below when no ambiguity can arise.

Assumption 2 (Design-measurable score regularity). *For every admissible partition Λ , the matrices $\mathbf{Q}_{K,\Lambda}$, $\mathbf{M}_{K,\Lambda}$, $\mathbf{\Pi}_{K,\Lambda} = \mathbf{I}_T - \mathbf{M}_{K,\Lambda}$, and $\mathbf{v}_\Lambda = \mathbf{M}_{K,\Lambda}\mathbf{w}$ are $\mathcal{D}_T$ -measurable. Hence the full-sample residualized instrument creates no predictability problem: all oracle score calculations are conditional-on-design calculations. For the true partition, write $\mathbf{v}_0 = \mathbf{v}_{\Lambda_0}$.*

Assumption 3 (Bounded weights, nuisance functions, and basis envelope). *The sieve dimension satisfies $K = K_T \rightarrow \infty$. The first basis element is constant, $p_1(w) \equiv 1$. There are constants $C_w, C_m < \infty$ and an envelope ζ_K such that*

$$\max_{t \leq T} |w_t| \leq C_w, \quad \max_{0 \leq j \leq q_0} \sup_x |m_j(x)| \leq C_m, \quad \max_{t \leq T} \|\mathbf{P}_K(W_{t-1})\| \leq \zeta_K$$

with probability tending to one.

Assumption 4 (Spacing, smoothness, and break detectability). *The known-partition theory allows fixed q_0 . The estimated-partition and unknown-order Wilks theorems in the main text impose $q_0 = 1$. The true segments obey the strict spacing condition in the setup with $\epsilon_0 > \epsilon$, and searches are over $\mathcal{L}_q(\epsilon)$ for $0 \leq q \leq \bar{q}$. Each m_j belongs to a common smoothness class $\mathcal{H}^s$. If $q_0 \geq 1$, define the design jump size by*

$$\Delta_T = \min_{0 \leq j < q_0} \left\{ T^{-1} \sum_{t=1}^T [m_{j+1}(W_{t-1}) - m_j(W_{t-1})]^2 \right\}^{1/2}.$$

For the one-break main theorem, this reduces to the empirical jump norm between m_0 and m_1 , and $\Delta_T > 0$.

Assumption 5 (Global and local Gram stability). *Uniformly over every interval J that can appear as a segment in an admissible search with $|J| \geq \epsilon T$,*

$$c_G I_K \preceq |J|^{-1} \mathbf{P}'_J \mathbf{P}_J \preceq C_G I_K$$

with probability tending to one. The same bounds hold on all one-break local segments satisfying $|k - k_0| \leq Cr_T$. This condition is stronger than algebraic Moore–Penrose existence: it supplies the inverse and leverage bounds used by the break bridge and by the profile-RSS search.

Assumption 6 (Sieve approximation and projection bias). *For each regime j , there is a coefficient vector $\mathbf{c}_{j,K}$ such that*

$$m_j(W_{t-1}) = \mathbf{P}_K(W_{t-1})' \mathbf{c}_{j,K} + r_{j,K,t}, \quad \max_j \|\mathbf{r}_{j,K}\|_T = O_p(a_K), \quad \max_j \|\mathbf{r}_{j,K}\|_\infty = O_p(a_K).$$

Let $\mathbf{r}_0$ be the stacked true-partition approximation error. The rates satisfy

$$T^{-1/2} \mathbf{r}'_0 \mathbf{M}_{K, \Lambda_0} \mathbf{w} = o_p(1), \quad a_K(1 + \zeta_K) \rightarrow 0, \quad a_K(1 + \zeta_K)^2 = o(\sqrt{T}).$$

Also,

$$\frac{(q_0 + 1)K}{\sqrt{T}} \rightarrow 0, \quad \zeta_K \sqrt{\frac{(q_0 + 1)K}{T}} \rightarrow 0.$$

Assumption 7 (Segmentwise orthogonality route for score bias). *For each true segment $I_{j,0}$, choose $\mathbf{c}_{j,K}$ as the empirical least-squares projection of $m_j(W_{t-1})$ on $\mathbf{P}_K(W_{t-1})$ within that segment, so*

$$\mathbf{P}'_{j,0} \mathbf{r}_{j,K} = 0.$$

The instrument admits a segmentwise sieve approximation

$$\mathbf{w}_{j,0} = \mathbf{P}_{j,0} \boldsymbol{\gamma}_{j,K} + \mathbf{s}_{j,K}, \quad \max_j \|\mathbf{s}_{j,K}\|_T = O_p(c_K),$$

where $\|\cdot\|_T$ uses the full-sample normalization. The rates satisfy

$$a_K \rightarrow 0, \quad \sqrt{T} a_K c_K \rightarrow 0.$$

Lemma 1 (Primitive score-bias verification). *Under Assumptions 5, 6 without its first score-bias display, and 7,*

$$T^{-1/2} \mathbf{r}'_0 \mathbf{M}_{K, \Lambda_0} \mathbf{w} = o_p(1).$$

Thus the score-bias display in Assumption 6 is discharged by segmentwise empirical projection and a smooth residualized instrument.

Detailed solution. On the true partition, $\mathbf{M}_{K, \Lambda_0}$ is block diagonal across the true segments. Let $\mathbf{M}_{j,0}$ be the residual-maker for $\mathbf{P}_{j,0}$ on segment j . Since $\mathbf{P}'_{j,0} \mathbf{r}_{j,K} = 0$, $\mathbf{M}_{j,0} \mathbf{r}_{j,K} = \mathbf{r}_{j,K}$. Also $\mathbf{M}_{j,0} \mathbf{w}_{j,0} = \mathbf{M}_{j,0} \mathbf{s}_{j,K}$. Hence

$$|\mathbf{r}'_0 \mathbf{M}_{K, \Lambda_0} \mathbf{w}| \leq \sum_{j=0}^{q_0} \|\mathbf{r}_{j,K}\| \|\mathbf{M}_{j,0} \mathbf{s}_{j,K}\| \leq \sum_{j=0}^{q_0} \|\mathbf{r}_{j,K}\| \|\mathbf{s}_{j,K}\|.$$

Because q_0 is fixed and the norms in Assumptions 6 and 7 use full-sample normalization, the last sum is $O_p(T a_K c_K)$. Multiplication by $T^{-1/2}$ gives $O_p(\sqrt{T} a_K c_K) = o_p(1)$. $\square$

Assumption 8 (One-break localization input for the bridge). *In the one-break bridge, $q_0 = 1$, $\Lambda_0 = (k_0)$, and $\tilde{\Lambda} = (\tilde{k})$. The bridge uses*

$$|\tilde{k} - k_0| = O_p(r_T), \quad r_T = o(\sqrt{T}).$$

Section 7 derives this rate from the conditional-design profile-sieve segmentation conditions with $r_T = R_T / \Delta_T^2$.

Assumption 9 (Empirical-likelihood convex hull). *For the score array under consideration,*

$$\mathbb{P}\{\min_{t \leq T} Z_{t,T} < 0 < \max_{t \leq T} Z_{t,T}\} \rightarrow 1.$$

When used for the true partition, $Z_{t,T} = \hat{Z}_t(\beta_0, \Lambda_0)$. When used for the estimated partition, $Z_{t,T} = \hat{Z}_t(\beta_0, \tilde{\Lambda})$.

Assumption 10 (Primitive sign mass for convex hull). *There are constants $c_u, c_v, p_0, \pi > 0$ and design-measurable sets $J_T \subset \{1, \dots, T\}$ with $|J_T| \geq \pi T$ such that*

$$|v_{t,0}| \geq c_v, \quad t \in J_T,$$

and, conditionally on $\mathcal{D}_T$,

$$\mathbb{P}(U_t > c_u \mid \mathcal{D}_T) \geq p_0, \quad \mathbb{P}(U_t < -c_u \mid \mathcal{D}_T) \geq p_0, \quad t \in J_T.$$

For the estimated one-break score, the feasible-score perturbation away from $O_p(r_T)$ boundary observations satisfies

$$a_K b_K^2 + b_K \zeta_K \sqrt{\frac{\log\{T(1+r_T)\}}{T}} + \frac{b_K r_T \zeta_K^2}{T} \rightarrow 0. \quad (2.1)$$

Lemma 2 (Primitive convex-hull verification). *Under Assumptions 1, 3, 5, 6, 11, and 10, Assumption 9 holds for the true partition score. If, in addition, $q_0 = 1$, $|\tilde{k} - k_0| = O_p(r_T)$, and the local bridge rates hold, then Assumption 9 also holds for the estimated one-break score.*

Detailed solution. Condition on $\mathcal{D}_T$. Let $J_T^+ = \{t \in J_T : v_{t,0} \geq c_v\}$ and $J_T^- = \{t \in J_T : v_{t,0} \leq -c_v\}$. At least one of these sets has cardinality at least $\pi T/2$. If $|J_T^+| \geq \pi T/2$, then a positive oracle score occurs when some $U_t > c_u$ on J_T^+ , and a negative oracle score occurs when some $U_t < -c_u$ on J_T^+ . Conditional independence bounds the probability of missing either sign by $2(1 - p_0)^{\pi T/2}$. The case $|J_T^-| \geq \pi T/2$ is identical with the signs reversed. For the true partition, the sharper leverage form of the projection-noise bound and the sieve approximation bound imply a uniform $o_p(1)$ feasible-score perturbation on J_T ; this is the $r_T = 0$ case of the transfer rate below. Thus both signs remain present for $\hat{Z}_t(\beta_0, \Lambda_0)$.

For the estimated partition, remove the boundary set where assignments differ; Lemma 10 leaves at least $\pi T - o_p(T)$ observations. On the remaining observations, the coefficient-update bound, the uniform projection-noise bound, and the sieve approximation bound give the transfer rate in (2.1). Hence the estimated score has both signs with probability tending to one. $\square$

3 Conditional-Design Completing Conditions

The rebuilt proof uses one stochastic environment throughout. Conditional on $\mathcal{D}_T$, the sieve design, all projection matrices, residualized weights, and break estimators are fixed. The remaining conditions below are rate and identification conditions; they do not reintroduce a predictable-score route.

Write

$$d_K = (q_0 + 1)K, \quad b_K = 1 + \zeta_K.$$

Assumption 11 (Oracle envelope completion). *For the true partition,*

$$\max_{t \leq T} |v_{t,0}| = O_p(b_K), \quad \frac{b_K^2 \log T}{T} \rightarrow 0.$$

The first display follows from the leverage bound under Assumptions 3 and 5; it is stated here to make the known-partition Wilks theorem self-contained. The logarithmic rate is the conditional sub-Gaussian maximum-score rate used below.

Assumption 12 (Local sieve-bias stability). *In the one-break section, for every fixed $C < \infty$,*

$$\sup_{|k - k_0| \leq Cr_T} T^{-1/2} |\mathbf{r}'_0 \{\mathbf{M}_{K,k} - \mathbf{M}_{K,k_0}\} \mathbf{w}| = o_p(1). \quad (3.1)$$

A sufficient primitive rate, proved by the coefficient-update argument in Lemma 12, is

$$\frac{a_K b_K (1 + \zeta_K^2) r_T}{\sqrt{T}} \rightarrow 0. \quad (3.2)$$

Assumption 13 (Local full-rank series regularity). *In the one-break section, every segment Gram matrix associated with a partition satisfying $|k - k_0| \leq Cr_T$ is nonsingular, and its eigenvalues are between cT and CT , uniformly with probability tending to one. Also*

$$\zeta_K^2 = O(K). \quad (3.3)$$

The last display is the usual normalized-series envelope condition. It is used to simplify the denominator bridge rate; without it the larger order $b_K^2 \zeta_K^2 r_T / T$ should be retained.

Assumption 14 (Deterministic conditional-design segmentation primitives). *This is the deterministic part of the route used to discharge the profile-sieve break search in the one-break main theorem. Let $\boldsymbol{\mu}_K = \mathbf{Q}_{K, \Lambda_0} \mathbf{c}_0$ be the sieve part of the true nuisance mean and define the deterministic conditional profile loss*

$$D_T(\Lambda) = \|\mathbf{M}_{K, \Lambda} \boldsymbol{\mu}_K\|^2.$$

At the true partition, $D_T(\Lambda_0) = 0$. For one-break candidates,

$$D_T(k) \geq c_\Delta |k - k_0| \Delta_T^2.$$

For underfitted candidates $q < q_0$,

$$\inf_{\Lambda \in \mathcal{L}_q(\epsilon)} D_T(\Lambda) \geq c_\Delta T \Delta_T^2.$$

The approximation error obeys the sup-norm rate in Assumption 6; write

$$R_T = K + \log T + T a_K^2, \quad r_T = R_T / \Delta_T^2.$$

The rates satisfy

$$R_T = o(T \Delta_T^2), \quad r_T = o(\sqrt{T}).$$

Together with Assumptions 1, 5, and 6, Proposition 1 verifies the profile-RSS localization and the overfit/underfit RSS gaps recorded in Assumption 15.

Remark 1 (Why the rebuild uses conditional design). The former martingale version needed a predictable approximation to the full-sample residualized instrument. The conditional-design route avoids that problem: $\mathbf{M}_{K, \Lambda}$, $\mathbf{v}_\Lambda$, and the RSS minimizer are all $\mathcal{D}_T$ -measurable, so stochastic arguments are conditional concentration and conditional triangular-array CLT arguments. The cost is a stronger exogeneity assumption and, in the main estimated-break Wilks theorem, the explicit one-break scope $q_0 = 1$.

Lemma 3 (Scalar empirical-likelihood expansion). *Let $Z_{1,T}, \dots, Z_{T,T}$ satisfy the convex-hull condition and suppose*

$$A_T = T^{-1/2} \sum_{t=1}^T Z_{t,T} = O_p(1), \quad B_T = T^{-1} \sum_{t=1}^T Z_{t,T}^2 \geq c + o_p(1)$$

for some $c > 0$, and

$$\max_{t \leq T} |Z_{t,T}| = o_p(\sqrt{T}).$$

Then

$$\ell_T(Z_{1,T}, \dots, Z_{T,T}) = \frac{A_T^2}{B_T} + o_p(1). \quad (3.4)$$

Proof. Put $S_1 = \sum_t Z_{t,T}$, $S_2 = \sum_t Z_{t,T}^2$, and $M_T = \max_t |Z_{t,T}|$. The convex-hull condition gives a unique multiplier λ_T in the interval on which every $1 + \lambda_T Z_{t,T}$ is positive. Its score equation implies the exact identity

$$S_1 = \lambda_T \sum_{t=1}^T \frac{Z_{t,T}^2}{1 + \lambda_T Z_{t,T}}.$$

Consequently,

$$|S_1| \geq \frac{|\lambda_T| S_2}{1 + |\lambda_T| M_T}.$$

After rearrangement,

$$|\lambda_T| \{S_2 - |S_1| M_T\} \leq |S_1|.$$

Now $|S_1| = O_p(\sqrt{T})$, $S_2 \geq (c + o_p(1))T$, and $|S_1| M_T = o_p(T)$. Hence

$$\lambda_T = O_p(T^{-1/2}), \quad |\lambda_T| M_T = o_p(1).$$

Using $(1+x)^{-1} = 1 - x + x^2/(1+x)$ in the score equation gives

$$0 = S_1 - \lambda_T S_2 + R_{1T}, \quad |R_{1T}| \leq \frac{\lambda_T^2 M_T S_2}{1 - |\lambda_T| M_T}.$$

Thus

$$\lambda_T = \frac{S_1}{S_2} + o_p(T^{-1/2}).$$

Finally, Taylor expansion of the logarithm, uniformly because $|\lambda_T| M_T = o_p(1)$, yields

$$2 \sum_t \log(1 + \lambda_T Z_{t,T}) = 2\lambda_T S_1 - \lambda_T^2 S_2 + R_{2T},$$

where

$$|R_{2T}| \leq C |\lambda_T|^3 M_T S_2 = o_p(1).$$

Substituting the expansion for λ_T gives

$$2 \sum_t \log(1 + \lambda_T Z_{t,T}) = \frac{S_1^2}{S_2} + o_p(1) = \frac{A_T^2}{B_T} + o_p(1),$$

which proves (3.4). □

4 Deterministic Projection Algebra

Lemma 4 (Block projection identities). *For every partition Λ ,*

$$\mathbf{Q}'_{K,\Lambda} \mathbf{M}_{K,\Lambda} = \mathbf{0}, \quad \mathbf{M}'_{K,\Lambda} = \mathbf{M}_{K,\Lambda}, \quad \mathbf{M}_{K,\Lambda}^2 = \mathbf{M}_{K,\Lambda}.$$

Consequently, for every conformable $\mathbf{c}$,

$$(\mathbf{Q}_{K,\Lambda} \mathbf{c})' \mathbf{M}_{K,\Lambda} \mathbf{w} = 0.$$

Detailed solution. Abbreviate $\mathbf{Q} = \mathbf{Q}_{K,\Lambda}$. Take a thin singular-value decomposition

$$\mathbf{Q} = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r',$$

where $r = \text{rank}(\mathbf{Q})$, the columns of $\mathbf{U}_r$ and $\mathbf{V}_r$ are orthonormal, and $\mathbf{\Sigma}_r$ is diagonal with strictly positive entries. Then

$$\mathbf{Q}'\mathbf{Q} = \mathbf{V}_r \mathbf{\Sigma}_r^2 \mathbf{V}_r', \quad (\mathbf{Q}'\mathbf{Q})^\dagger = \mathbf{V}_r \mathbf{\Sigma}_r^{-2} \mathbf{V}_r'.$$

Therefore

$$\mathbf{\Pi}_{K,\Lambda} = \mathbf{Q}(\mathbf{Q}'\mathbf{Q})^\dagger \mathbf{Q}' = \mathbf{U}_r \mathbf{U}_r'.$$

This is the orthogonal projector onto $\text{col}(\mathbf{Q})$. In particular,

$$\mathbf{\Pi}_{K,\Lambda}' = \mathbf{\Pi}_{K,\Lambda}, \quad \mathbf{\Pi}_{K,\Lambda}^2 = \mathbf{\Pi}_{K,\Lambda}, \quad \mathbf{\Pi}_{K,\Lambda} \mathbf{Q} = \mathbf{Q}.$$

Since $\mathbf{M}_{K,\Lambda} = \mathbf{I}_T - \mathbf{\Pi}_{K,\Lambda}$,

$$\mathbf{M}_{K,\Lambda}' = \mathbf{M}_{K,\Lambda}$$

and

$$\mathbf{M}_{K,\Lambda}^2 = (\mathbf{I}_T - \mathbf{\Pi}_{K,\Lambda})^2 = \mathbf{I}_T - 2\mathbf{\Pi}_{K,\Lambda} + \mathbf{\Pi}_{K,\Lambda}^2 = \mathbf{M}_{K,\Lambda}.$$

Also

$$\mathbf{Q}'\mathbf{M}_{K,\Lambda} = \{\mathbf{M}_{K,\Lambda}\mathbf{Q}\}' = \mathbf{0}'.$$

For conformable $\mathbf{c}$ and $\mathbf{w}$, it follows immediately that

$$(\mathbf{Q}\mathbf{c})'\mathbf{M}_{K,\Lambda}\mathbf{w} = \mathbf{c}'\mathbf{Q}'\mathbf{M}_{K,\Lambda}\mathbf{w} = 0.$$

□

Lemma 5 (Uniform fitted-value and leverage bounds). *Under Assumptions 3 and 5, uniformly over*

$$\mathcal{N}_T(r_T) = \{\Lambda : q = q_0, \max_{1 \leq j \leq q_0} |k_j - k_{j,0}| \leq Cr_T\},$$

$$\|\mathbf{M}_{K,\Lambda}\mathbf{w}\|_\infty = O_p(1 + \zeta_K), \quad \max_{t \leq T} \{\mathbf{Q}_{K,\Lambda}(\mathbf{Q}_{K,\Lambda}'\mathbf{Q}_{K,\Lambda})^\dagger \mathbf{Q}_{K,\Lambda}'\}_{tt} = O_p(\zeta_K^2 K/T).$$

Detailed solution. Write $\mathbf{Q} = \mathbf{Q}_{K,\Lambda}$, $\mathbf{G} = \mathbf{Q}'\mathbf{Q}$, and let $\mathbf{q}_t'$ denote row t of $\mathbf{Q}$. Because exactly one regime block is active in each row,

$$\|\mathbf{q}_t\| = \|\mathbf{P}_K(\mathbf{W}_{t-1})\| \leq \zeta_K.$$

On the uniform Gram event in Assumption 5,

$$\|\mathbf{G}^\dagger\| \leq \frac{C}{T}, \quad \|\mathbf{Q}\|^2 = \lambda_{\max}(\mathbf{Q}'\mathbf{Q}) \leq CT.$$

Since $\|\mathbf{w}\| \leq C_w \sqrt{T}$, the least-squares coefficient vector

$$\hat{\mathbf{c}}_\Lambda = \mathbf{G}^\dagger \mathbf{Q}'\mathbf{w}$$

satisfies

$$\|\widehat{\mathbf{c}}_\Lambda\| \leq \frac{C}{T} \|\mathbf{Q}\| \|\mathbf{w}\| \leq C.$$

Thus, uniformly in t and Λ ,

$$|(\mathbf{\Pi}_{K,\Lambda}\mathbf{w})_t| = |\mathbf{q}'_t \widehat{\mathbf{c}}_\Lambda| \leq C\zeta_K,$$

and hence

$$\|\mathbf{M}_{K,\Lambda}\mathbf{w}\|_\infty \leq \|\mathbf{w}\|_\infty + \|\mathbf{\Pi}_{K,\Lambda}\mathbf{w}\|_\infty = O_p(1 + \zeta_K).$$

For the leverage,

$$(\mathbf{\Pi}_{K,\Lambda})_{tt} = \mathbf{q}'_t \mathbf{G}^\dagger \mathbf{q}_t \leq \|\mathbf{G}^\dagger\| \|\mathbf{q}_t\|^2 \leq C \frac{\zeta_K^2}{T}.$$

This is sharper than the displayed target and therefore implies $O_p(\zeta_K^2 K/T)$, because $K \geq 1$. All inequalities hold on a single uniform Gram event whose probability tends to one. $\square$

5 Known-Partition Conditional-Design Wilks

Set

$$\mathbf{v}_0 = \mathbf{M}_{K,\Lambda_0}\mathbf{w}, \quad A_T^{(q)} = T^{-1/2} \sum_{t=1}^T U_t v_{t,0}, \quad V_T^{(q)} = T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2.$$

Lemma 6 (Oracle conditional-design CLT). *Under Assumptions 1–5 and 11, suppose*

$$\eta_{T,q}^2 := T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2, \quad \mathbb{P}(\eta_q \geq \sqrt{c_\eta}) = 1$$

for some $c_\eta > 0$. Then

$$A_T^{(q)} \xrightarrow{\text{st}} \eta_q Z,$$

where $Z \sim N(0, 1)$ is independent of the limiting sigma-field generated by η_q .

Detailed solution. Condition on $\mathcal{D}_T$. The summands $T^{-1/2}U_t v_{t,0}$ are independent, centered, and have conditional variance $T^{-1}\sigma_t^2 v_{t,0}^2$. The conditional variance sum is $\eta_{T,q}^2$. The conditional sub-Gaussian envelope and $\max_t |v_{t,0}| = O_p(b_K)$, together with $b_K/\sqrt{T} \rightarrow 0$, imply the Lindeberg condition conditionally on $\mathcal{D}_T$: for every $\varepsilon > 0$,

$$\sum_{t=1}^T \mathbb{E} \left[T^{-1} U_t^2 v_{t,0}^2 \mathbf{1}\{|U_t v_{t,0}| > \varepsilon \sqrt{T}\} \mid \mathcal{D}_T \right] = o_p(1).$$

Therefore the conditional triangular-array CLT gives

$$\mathbb{E}\{\exp(iu A_T^{(q)}) \mid \mathcal{D}_T\} - \exp(-u^2 \eta_{T,q}^2/2) \xrightarrow{p} 0$$

for every fixed u . Since $\eta_{T,q}^2 \rightarrow_p \eta_q^2$, the usual conditional-characteristic-function argument yields stable convergence to $\eta_q Z$. The lower bound on η_q is recorded for the later studentization step. $\square$

Lemma 7 (Oracle denominator LLN). *Under Assumptions 1–5 and 11, suppose*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2.$$

Then

$$V_T^{(q)} - \eta_q^2 = o_p(1).$$

Detailed solution. Conditional on $\mathcal{D}_T$, the variables $(U_t^2 - \sigma_t^2)v_{t,0}^2$ are independent and centered. Conditional sub-Gaussianity gives a uniform fourth-moment bound, hence

$$\text{Var} \left(T^{-1} \sum_{t=1}^T (U_t^2 - \sigma_t^2)v_{t,0}^2 \mid \mathcal{D}_T \right) \leq CT^{-2} \sum_{t=1}^T v_{t,0}^4.$$

Using $\max_t |v_{t,0}| = O_p(b_K)$ and $T^{-1} \sum_t v_{t,0}^2 = O_p(1)$, the right side is $O_p(b_K^2/T) = o_p(1)$. Thus

$$T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2 - T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 = o_p(1),$$

and the displayed variance convergence proves the lemma. $\square$

Lemma 8 (Oracle maximum bound). *Under Assumptions 1, 3, 5, and 11,*

$$\max_{t \leq T} |U_t v_{t,0}| = o_p(\sqrt{T}).$$

Detailed solution. Conditional on $\mathcal{D}_T$, sub-Gaussian maximal inequalities give

$$\max_{t \leq T} |U_t| = O_p(\sqrt{\log T}).$$

Lemma 5 and Assumption 11 give $\max_t |v_{t,0}| = O_p(b_K)$. Therefore

$$\max_{t \leq T} |U_t v_{t,0}| = O_p(b_K \sqrt{\log T}) = o_p(\sqrt{T}),$$

by $b_K^2 \log T/T \rightarrow 0$. $\square$

Lemma 9 (Feasible true-partition replacement). *Under Assumptions 1–6 and 11, suppose*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2.$$

Then

$$T^{-1/2} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^T U_t v_{t,0} + o_p(1), \quad (5.1)$$

$$T^{-1} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0)^2 = T^{-1} \sum_{t=1}^T U_t^2 v_{t,0}^2 + o_p(1), \quad (5.2)$$

$$\max_{t \leq T} |\widehat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}). \quad (5.3)$$

Detailed solution. Let $\mathbf{Q}_0 = \mathbf{Q}_{K,\Lambda_0}$, $\mathbf{\Pi}_0 = \mathbf{I}_T - \mathbf{M}_{K,\Lambda_0}$, and $\mathbf{v}_0 = \mathbf{M}_{K,\Lambda_0}\mathbf{w}$. The sieve representation gives

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \mathbf{Q}_0 \mathbf{c}_0 + \mathbf{r}_0 + \mathbf{U}.$$

By Lemma 4,

$$\hat{\mathbf{u}}(\beta_0, \Lambda_0) = \mathbf{M}_{K,\Lambda_0}(\mathbf{r}_0 + \mathbf{U}).$$

Define

$$\mathbf{e}_0 = \mathbf{M}_{K,\Lambda_0} \mathbf{r}_0, \quad \mathbf{h}_0 = \mathbf{\Pi}_0 \mathbf{U}, \quad \mathbf{d}_0 = (\mathbf{e}_0 - \mathbf{h}_0) \odot \mathbf{v}_0.$$

Then

$$\hat{Z}_t(\beta_0, \Lambda_0) = U_t v_{t,0} + d_{t,0}.$$

Conditional on $\mathcal{D}_T$, $\mathbf{Q}'_0 \mathbf{U}$ is a sub-Gaussian vector with second moment of order $T d_K$. Hence

$$\|\mathbf{Q}'_0 \mathbf{U}\| = O_p(\sqrt{T d_K}).$$

The Gram bounds imply

$$\|\mathbf{h}_0\|_T = O_p(\sqrt{d_K/T}), \quad \|\mathbf{h}_0\|_\infty = O_p(\zeta_K \sqrt{d_K/T}).$$

Projection contraction and Assumption 6 give

$$\|\mathbf{e}_0\|_T = O_p(a_K), \quad \|\mathbf{e}_0\|_\infty = O_p\{a_K(1 + \zeta_K)\}.$$

Since $\|\mathbf{v}_0\|_\infty = O_p(b_K)$,

$$\|\mathbf{d}_0\|_T = O_p(a_K b_K + b_K \sqrt{d_K/T}) = o_p(1),$$

and

$$\|\mathbf{d}_0\|_\infty = O_p(a_K b_K^2 + \zeta_K b_K \sqrt{d_K/T}) = o_p(\sqrt{T}).$$

The numerator has the exact cancellation

$$T^{-1/2} \sum_t \hat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \mathbf{U}' \mathbf{v}_0 + T^{-1/2} \mathbf{r}'_0 \mathbf{M}_{K,\Lambda_0} \mathbf{w} = T^{-1/2} \mathbf{U}' \mathbf{v}_0 + o_p(1).$$

For the denominator,

$$\left| T^{-1} \sum_t \hat{Z}_t^2 - T^{-1} \sum_t U_t^2 v_{t,0}^2 \right| \leq 2 \|\mathbf{U} \odot \mathbf{v}_0\|_T \|\mathbf{d}_0\|_T + \|\mathbf{d}_0\|_T^2 = o_p(1),$$

because Lemma 7 gives $\|\mathbf{U} \odot \mathbf{v}_0\|_T = O_p(1)$. The maximum bound follows from Lemma 8 and $\|\mathbf{d}_0\|_\infty = o_p(\sqrt{T})$. $\square$

Theorem 1 (Known-partition conditional-design Wilks). *Under Assumptions 1–6, 9, and 11, suppose*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2, \quad \mathbb{P}(\eta_q \geq \sqrt{c_\eta}) = 1$$

for some $c_\eta > 0$. Then

$$\ell_T(\beta_0, \Lambda_0) \Rightarrow \chi_1^2.$$

Detailed solution. Let

$$A_T = T^{-1/2} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0), \quad B_T = T^{-1} \sum_{t=1}^T \widehat{Z}_t(\beta_0, \Lambda_0)^2.$$

Lemma 9, followed by Lemmas 6 and 7, gives

$$A_T \xrightarrow{\text{st}} \eta_q Z, \quad B_T \xrightarrow{p} \eta_q^2.$$

Because $\eta_q^2 \geq c_\eta$ almost surely, $B_T \geq c_\eta/2$ with probability tending to one. Lemmas 8 and 9 give

$$\max_t |\widehat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}).$$

Assumption 9 supplies the convex-hull condition. Lemma 3 gives

$$\ell_T(\beta_0, \Lambda_0) = A_T^2/B_T + o_p(1),$$

and stable Slutsky yields $A_T^2/B_T \Rightarrow Z^2 \sim \chi_1^2$. □

Remark 2 (Proof audit for Theorem 1). The known-partition result now uses a single stochastic route. Conditional on $\mathcal{D}_T$, the residualized score weights are fixed, so the numerator CLT, denominator LLN, and maximum bound are ordinary conditional independent-array arguments. Break-date localization and order selection are irrelevant until Λ_0 is replaced by an estimated partition.

6 One-Break Estimated-Partition Bridge

In this section only, take $q_0 = 1$, $\Lambda_0 = (k_0)$, and $\tilde{\Lambda} = (\tilde{k})$. Let

$$\Delta_M = \mathbf{M}_{K, \tilde{\Lambda}} - \mathbf{M}_{K, \Lambda_0}, \quad \mathcal{M}_T = \{t : j_{\tilde{\Lambda}}(t) \neq j_{\Lambda_0}(t)\}.$$

Lemma 10 (Boundary-set size). *If $|\tilde{k} - k_0| = O_p(r_T)$, then*

$$|\mathcal{M}_T| = O_p(r_T).$$

Detailed solution. Suppose first that $\tilde{k} \geq k_0$. For $t \leq k_0$, both partitions assign observation t to the left regime; for $t > \tilde{k}$, both assign it to the right regime. The assignments differ exactly on

$$k_0 < t \leq \tilde{k}.$$

Thus $|\mathcal{M}_T| = \tilde{k} - k_0$. If $\tilde{k} < k_0$, the same argument gives the differing set $\tilde{k} < t \leq k_0$, so $|\mathcal{M}_T| = k_0 - \tilde{k}$. Hence, in all cases,

$$|\mathcal{M}_T| = |\tilde{k} - k_0| = O_p(r_T).$$

□

Lemma 11 (Search-uniform projection noise). *Suppose $q_0 = 1$ and Assumptions 1, 3, 5, and 13 hold. For every fixed $C < \infty$, over the local one-break set*

$$\mathcal{N}_T(Cr_T) = \{k : |k - k_0| \leq Cr_T\},$$

$$\sup_{k \in \mathcal{N}_T(Cr_T)} \|\mathbf{\Pi}_k \mathbf{U}\|_T = O_p \left(\sqrt{\frac{K + \log(1 + r_T)}{T}} \right),$$

and

$$\sup_{k \in \mathcal{N}_T(Cr_T)} \|\mathbf{\Pi}_k \mathbf{U}\|_\infty = O_p \left(\zeta_K \sqrt{\frac{\log\{T(1 + r_T)\}}{T}} \right).$$

Detailed solution. Condition on $\mathcal{D}_T$. For fixed k , $\mathbf{\Pi}_k \mathbf{U}$ is the projection of a conditionally sub-Gaussian vector onto a subspace of rank at most $2K$. Standard chi-square/sub-Gaussian projection concentration gives

$$\|\mathbf{\Pi}_k \mathbf{U}\|^2 = O_p(K + x)$$

with conditional tail probability bounded by e^{-cx} . A union bound over $O(1 + r_T)$ local candidates yields the displayed T -norm bound. For the sup norm, the leverage bound gives $(\mathbf{\Pi}_k)_{tt} \leq C\zeta_K^2/T$ uniformly over t and local k . Conditional sub-Gaussian tails for each coordinate, followed by a union bound over $T(1 + r_T)$ pairs (t, k) , give the displayed sup-norm bound. $\square$

Lemma 12 (Nuisance part of partition perturbation). *Suppose $q_0 = 1$, Assumptions 3, 5, 6, and 12 hold, and $|\tilde{k} - k_0| = O_p(r_T)$. Then*

$$T^{-1/2} \boldsymbol{\mu}' \Delta_M \mathbf{w} = O_p \left(\frac{b_K r_T}{\sqrt{T}} \right) + o_p(1), \quad b_K = 1 + \zeta_K.$$

In particular, this term is $o_p(1)$ if $b_K r_T / \sqrt{T} \rightarrow 0$.

Detailed solution. It suffices to prove a uniform deterministic bound for all $|k - k_0| \leq Cr_T$, then evaluate it at $k = \tilde{k}$ on the localization event. Consider first $k = k_0 + h$, $h \geq 0$, and let $H = \{k_0 + 1, \dots, k_0 + h\}$. Let $\mathbf{c}_0$ and $\mathbf{c}_1$ be the true-regime sieve coefficients. If the candidate partition uses $\mathbf{c}_0$ on its left block and $\mathbf{c}_1$ on its right block, then

$$\boldsymbol{\mu} = \mathbf{Q}_{K,k} \mathbf{c}^{(k)} + \mathbf{d}_H + \mathbf{r}_0,$$

where $\mathbf{d}_H$ is supported on H . For $t \in H$,

$$d_{H,t} = \mathbf{P}_K(W_{t-1})'(\mathbf{c}_1 - \mathbf{c}_0) = m_1(W_{t-1}) - m_0(W_{t-1}) - \{r_{1,K,t} - r_{0,K,t}\}.$$

Assumptions 3 and 6 imply $\|\mathbf{d}_H\|_\infty = O_p(1)$. Since $\mathbf{Q}'_{K,k} \mathbf{M}_{K,k} = \mathbf{0}$,

$$\boldsymbol{\mu}' \mathbf{M}_{K,k} \mathbf{w} = \mathbf{d}_H' \mathbf{M}_{K,k} \mathbf{w} + \mathbf{r}_0' \mathbf{M}_{K,k} \mathbf{w}.$$

At the true partition,

$$\boldsymbol{\mu}' \mathbf{M}_{K,k_0} \mathbf{w} = \mathbf{r}_0' \mathbf{M}_{K,k_0} \mathbf{w}.$$

Therefore, using Lemma 5 uniformly over local candidates,

$$\begin{aligned} T^{-1/2}|\boldsymbol{\mu}'(\mathbf{M}_{K,k} - \mathbf{M}_{K,k_0})\mathbf{w}| &\leq T^{-1/2}\|\mathbf{d}_H\|_1\|\mathbf{M}_{K,k}\mathbf{w}\|_\infty \\ &\quad + T^{-1/2}|\mathbf{r}'_0(\mathbf{M}_{K,k} - \mathbf{M}_{K,k_0})\mathbf{w}| \\ &= O_p\left(\frac{b_K h}{\sqrt{T}}\right) + o_p(1), \end{aligned}$$

where the final term is Assumption 12. The case $k < k_0$ is identical after exchanging the left and right regimes. Since $|\tilde{k} - k_0| = O_p(r_T)$, the lemma follows.

The sufficient primitive rate in (3.2) follows from the same local coefficient update. Write $\mathbf{v}_k = \mathbf{M}_{K,k}\mathbf{w}$. On the boundary set H , $|v_{t,k} - v_{t,k_0}| = O_p(b_K)$. Off the boundary, the candidate and true fitted-weight coefficients differ by $O_p(h\zeta_K b_K/T)$, and each row has norm at most ζ_K . Hence

$$\|\mathbf{v}_k - \mathbf{v}_{k_0}\|_1 = O_p\{hb_K(1 + \zeta_K^2)\}.$$

Consequently,

$$T^{-1/2}|\mathbf{r}'_0(\mathbf{v}_k - \mathbf{v}_{k_0})| \leq T^{-1/2}\|\mathbf{r}_0\|_\infty\|\mathbf{v}_k - \mathbf{v}_{k_0}\|_1 = O_p\left(\frac{a_K b_K(1 + \zeta_K^2)h}{\sqrt{T}}\right),$$

which is uniformly $o_p(1)$ for $h \leq Cr_T$ under (3.2). $\square$

Lemma 13 (Stochastic part of partition perturbation). *Suppose $q_0 = 1$, Assumptions 1, 3, 5, and 13 hold, $|\tilde{k} - k_0| = O_p(r_T)$, and $b_K r_T/\sqrt{T} \rightarrow 0$. Then*

$$T^{-1/2}\mathbf{U}'\Delta_M\mathbf{w} = O_p\left(\zeta_K\sqrt{\frac{r_T\{K + \log(1 + r_T)\}}{T}}\right) + o_p(1).$$

In particular, this term is $o_p(1)$ if $\zeta_K\sqrt{r_T\{K + \log(1 + r_T)\}}/T \rightarrow 0$.

Detailed solution. Work uniformly over deterministic local candidates $|k - k_0| \leq Cr_T$, then evaluate at $k = \tilde{k}$ on the localization event. Take $k = k_0 + h > k_0$ and define

$$A = \{1, \dots, k_0\}, \quad H = \{k_0 + 1, \dots, k_0 + h\}, \quad B = \{k_0 + h + 1, \dots, T\}, \quad R = H \cup B.$$

For a segment J , put

$$\mathbf{G}_J = \mathbf{P}'_J \mathbf{P}_J, \quad \hat{\boldsymbol{\gamma}}_J = \mathbf{G}_J^{-1} \mathbf{P}'_J \mathbf{w}_J.$$

Assumption 13 gives $\|\mathbf{G}_J^{-1}\| = O_p(T^{-1})$ for all long local segments, and bounded weights imply $\|\hat{\boldsymbol{\gamma}}_J\| = O_p(1)$. The exact update identity

$$\hat{\boldsymbol{\gamma}}_{A \cup H} - \hat{\boldsymbol{\gamma}}_A = \mathbf{G}_{A \cup H}^{-1} \mathbf{P}'_H (\mathbf{w}_H - \mathbf{P}_H \hat{\boldsymbol{\gamma}}_A)$$

therefore yields

$$\|\hat{\boldsymbol{\gamma}}_{A \cup H} - \hat{\boldsymbol{\gamma}}_A\| = O_p\left(\frac{h\zeta_K b_K}{T}\right).$$

The same calculation after removing H from the true right segment gives

$$\|\hat{\gamma}_B - \hat{\gamma}_R\| = O_p\left(\frac{h\zeta_K b_K}{T}\right).$$

The residualized weights are segmentwise

$$\begin{aligned} \mathbf{v}_{k_0} &= (\mathbf{w}_A - \mathbf{P}_A \hat{\gamma}_A, \mathbf{w}_R - \mathbf{P}_R \hat{\gamma}_R), \\ \mathbf{v}_k &= (\mathbf{w}_{A \cup H} - \mathbf{P}_{A \cup H} \hat{\gamma}_{A \cup H}, \mathbf{w}_B - \mathbf{P}_B \hat{\gamma}_B). \end{aligned}$$

Cancelling the raw $\mathbf{w}_H$ terms gives the exact decomposition

$$\begin{aligned} \mathbf{U}'(\mathbf{v}_k - \mathbf{v}_{k_0}) &= (\mathbf{P}'_A \mathbf{U}_A)'(\hat{\gamma}_A - \hat{\gamma}_{A \cup H}) \\ &\quad + (\mathbf{P}'_B \mathbf{U}_B)'(\hat{\gamma}_R - \hat{\gamma}_B) \\ &\quad + (\mathbf{P}'_H \mathbf{U}_H)'(\hat{\gamma}_R - \hat{\gamma}_{A \cup H}). \end{aligned} \tag{6.1}$$

Conditional on $\mathcal{D}_T$, sub-Gaussian concentration and the Gram bounds give

$$\|\mathbf{P}'_A \mathbf{U}_A\| = O_p(\sqrt{TK}), \quad \|\mathbf{P}'_B \mathbf{U}_B\| = O_p(\sqrt{TK}).$$

For the local moving block, a union bound over $h \leq Cr_T$ gives

$$\sup_{h \leq Cr_T} \|\mathbf{P}'_H \mathbf{U}_H\| = O_p\{\zeta_K \sqrt{r_T \log(1 + r_T)}\}.$$

The coefficient difference in the last line of (6.1) is $O_p(1)$. Combining the displays, uniformly for $h \leq Cr_T$,

$$T^{-1/2} |\mathbf{U}'(\mathbf{v}_k - \mathbf{v}_{k_0})| = O_p\left(\frac{h\zeta_K b_K \sqrt{K}}{T}\right) + O_p\left(\zeta_K \sqrt{\frac{r_T \log(1 + r_T)}{T}}\right).$$

The first order is negligible under $b_K r_T / \sqrt{T} \rightarrow 0$, and the second order is exactly the logarithmic local-search term. Taking $h = O_p(r_T)$ proves the displayed result. The case $k < k_0$ is the same with left and right interchanged. $\square$

Lemma 14 (Denominator perturbation). *Suppose $q_0 = 1$, Assumptions 1, 2, 3, 5, 6, 11, 12, and 13 hold. Suppose also*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2, \quad |\tilde{k} - k_0| = O_p(r_T),$$

and

$$\frac{b_K r_T}{\sqrt{T}} \rightarrow 0, \quad b_K \sqrt{\frac{K + \log(1 + r_T)}{T}} \rightarrow 0, \quad \frac{\zeta_K^2 K r_T}{T} \rightarrow 0.$$

Then

$$V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = O_p\left(\frac{b_K^2 r_T}{T} + \frac{\zeta_K^2 K r_T}{T}\right) + o_p(1) = o_p(1).$$

Detailed solution. Again work uniformly over $|k - k_0| \leq Cr_T$. Put $\mathbf{v}_k = \mathbf{M}_{K,k}\mathbf{w}$ and $V_{U,T}(k) = T^{-1} \sum_t U_t^2 v_{t,k}^2$. As in Lemma 12, for a local candidate partition,

$$\boldsymbol{\mu} = \mathbf{Q}_{K,k}\mathbf{c}^{(k)} + \mathbf{d}_H + \mathbf{r}_0,$$

where $\mathbf{d}_H$ is supported on the boundary set and $\|\mathbf{d}_H\|_\infty = O_p(1)$. Therefore

$$\hat{\mathbf{u}}(\beta_0, k) = \mathbf{U} - \boldsymbol{\Pi}_k \mathbf{U} + \mathbf{M}_{K,k} \mathbf{r}_0 + \mathbf{M}_{K,k} \mathbf{d}_H, \quad \boldsymbol{\Pi}_k = \mathbf{I}_T - \mathbf{M}_{K,k},$$

and

$$\hat{\mathbf{Z}}(\beta_0, k) = \mathbf{U} \odot \mathbf{v}_k + \boldsymbol{\rho}_k,$$

with

$$\|\boldsymbol{\rho}_k\|_T \leq \|\mathbf{v}_k\|_\infty \{\|\boldsymbol{\Pi}_k \mathbf{U}\|_T + \|\mathbf{M}_{K,k} \mathbf{r}_0\|_T + \|\mathbf{M}_{K,k} \mathbf{d}_H\|_T\}.$$

Lemma 11 gives, uniformly over the local full-rank set,

$$\|\boldsymbol{\Pi}_k \mathbf{U}\|_T = O_p \left(\sqrt{\frac{K + \log(1 + r_T)}{T}} \right).$$

Projection contraction gives

$$\|\mathbf{M}_{K,k} \mathbf{r}_0\|_T = O_p(a_K), \quad \|\mathbf{M}_{K,k} \mathbf{d}_H\|_T = O_p(\sqrt{h/T}).$$

Since $\|\mathbf{v}_k\|_\infty = O_p(b_K)$, Assumptions 6 and 11, together with $b_K r_T / \sqrt{T} \rightarrow 0$, imply

$$\sup_{|k - k_0| \leq Cr_T} \|\boldsymbol{\rho}_k\|_T = o_p(1). \quad (6.2)$$

It remains to compare the oracle denominators. The coefficient-update bounds from Lemma 13 imply, for $h = |k - k_0|$,

$$|v_{t,k} - v_{t,k_0}| \leq \begin{cases} Cb_K, & t \in \mathcal{M}_T, \\ Ch\zeta_K^2 b_K / T, & t \notin \mathcal{M}_T. \end{cases}$$

Also $\max_t (|v_{t,k}| + |v_{t,k_0}|) = O_p(b_K)$. For a boundary interval of length h , conditional sub-Gaussian concentration and a dyadic union bound give

$$\sup_{1 \leq h \leq Cr_T} \frac{1}{h} \sum_{t \in H_h} U_t^2 = O_p(1),$$

and $T^{-1} \sum_t U_t^2 = O_p(1)$. Hence, uniformly over local candidates,

$$\begin{aligned} |V_{U,T}(k) - V_{U,T}(k_0)| &\leq T^{-1} \sum_t U_t^2 |v_{t,k} - v_{t,k_0}| (|v_{t,k}| + |v_{t,k_0}|) \\ &= O_p \left(\frac{b_K^2 h}{T} \right) + O_p \left(\frac{h \zeta_K^2 b_K^2}{T} \right). \end{aligned}$$

Assumption 13 gives $b_K^2 = O(K)$, so the second term is $O_p(\zeta_K^2 Kh/T)$. Thus

$$|V_{U,T}(k) - V_{U,T}(k_0)| = O_p\left(\frac{b_K^2 h}{T} + \frac{\zeta_K^2 Kh}{T}\right).$$

By Lemma 7, $V_{U,T}(k_0) = O_p(1)$; hence $\|\mathbf{U} \odot \mathbf{v}_k\|_T = O_p(1)$ uniformly locally. The square-difference inequality and (6.2) imply

$$V_T(k) - V_{U,T}(k) = o_p(1)$$

uniformly locally, and the same is true at k_0 . Taking $h = |\tilde{k} - k_0| = O_p(r_T)$ gives the displayed bound. $\square$

Theorem 2 (One-break estimated-partition score equivalence). *Suppose $q_0 = 1$, Assumptions 1–8, 11, 12, and 13 hold, and*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2.$$

Assume $r_T \geq 1$ and the bridge rates

$$\zeta_K \sqrt{\frac{r_T \{K + \log(1 + r_T)\}}{T}} \rightarrow 0, \quad \frac{b_K r_T}{\sqrt{T}} \rightarrow 0, \quad \frac{\zeta_K^2 K r_T}{T} \rightarrow 0.$$

Then

$$S_T(\tilde{\Lambda}) - S_T(\Lambda_0) = o_p(1), \quad V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = o_p(1).$$

Detailed solution. Assumption 8 gives $|\tilde{k} - k_0| = O_p(r_T)$. For every one-break partition Λ , symmetry and idempotence imply

$$S_T(\Lambda) = T^{-1/2}(\mathbf{U} + \boldsymbol{\mu})' \mathbf{M}_{K,\Lambda} \mathbf{w}.$$

Therefore

$$S_T(\tilde{\Lambda}) - S_T(\Lambda_0) = T^{-1/2} \boldsymbol{\mu}' \Delta_M \mathbf{w} + T^{-1/2} \mathbf{U}' \Delta_M \mathbf{w}.$$

Lemma 12 bounds the nuisance term by

$$O_p\left(\frac{b_K r_T}{\sqrt{T}}\right) + o_p(1) = o_p(1),$$

and Lemma 13 bounds the stochastic term by

$$O_p\left(\zeta_K \sqrt{\frac{r_T \{K + \log(1 + r_T)\}}{T}}\right) + o_p(1) = o_p(1).$$

This proves the numerator equivalence.

For the denominator, Lemma 14 gives

$$V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = O_p\left(\frac{b_K^2 r_T}{T} + \frac{\zeta_K^2 K r_T}{T}\right) + o_p(1).$$

The third bridge rate makes the second term $o_p(1)$. Since $r_T \geq 1$,

$$\frac{b_K^2 r_T}{T} = \frac{1}{r_T} \left(\frac{b_K r_T}{\sqrt{T}} \right)^2 = o(1),$$

so the denominator difference is $o_p(1)$. $\square$

Remark 3 (Proof audit for Theorem 2). Theorem 2 is only a bridge from the true one-break partition to a localized estimated break. Lemma 10 limits the number of observations whose regime assignment can change. Lemma 12 handles the deterministic nuisance component and makes the local sieve-bias condition explicit. Lemma 13 handles the stochastic component through conditional sub-Gaussian concentration and segmentwise coefficient updates. Lemma 14 proves that the empirical second moment is stable after the same partition perturbation. No convex-hull condition and no empirical-likelihood expansion is used in this theorem; those enter only in the next Wilks theorem. The result would fail without localization, local full-rank Gram regularity, and the three bridge rates listed in the theorem statement.

Theorem 3 (One-break estimated-partition Wilks). *Under the assumptions of Theorem 1, the convex-hull condition for the estimated-partition score array, and Theorem 2,*

$$\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2.$$

Detailed solution. Let $A_T(\Lambda) = S_T(\Lambda)$ and $B_T(\Lambda) = V_T(\Lambda)$. Theorem 2 gives

$$A_T(\tilde{\Lambda}) = A_T(\Lambda_0) + o_p(1), \quad B_T(\tilde{\Lambda}) = B_T(\Lambda_0) + o_p(1).$$

The known-partition proof gives

$$A_T(\Lambda_0) \xrightarrow{\text{st}} \eta_q Z, \quad B_T(\Lambda_0) \xrightarrow{p} \eta_q^2.$$

We must also verify the maximum condition; numerator and denominator equivalence alone would not imply it. The local representation in the proof of Lemma 14 gives

$$\hat{\mathbf{Z}}(\beta_0, \tilde{\Lambda}) = \mathbf{U} \odot \mathbf{v}_{\tilde{k}} + \boldsymbol{\rho}_{\tilde{k}}.$$

The envelope rate implies $\max_t |U_t v_{t,\tilde{k}}| = o_p(\sqrt{T})$. For the remainder, the uniform projection calculation gives

$$\|\boldsymbol{\Pi}_{\tilde{k}} \mathbf{U}\|_\infty = O_p \left(\zeta_K \sqrt{\frac{\log\{T(1+r_T)\}}{T}} \right),$$

while the sieve calculation gives

$$\|\mathbf{M}_{K,\tilde{k}} \mathbf{r}_0\|_\infty = O_p\{a_K(1 + \zeta_K)\}.$$

Finally, because $\mathbf{d}_H$ has bounded entries and support of size $O_p(r_T)$,

$$\|\mathbf{M}_{K,\tilde{k}} \mathbf{d}_H\|_\infty = O_p \left(1 + \frac{r_T \zeta_K^2}{T} \right) = O_p(1).$$

Multiplication by $\|\mathbf{v}_{\tilde{k}}\|_\infty = O_p(b_K)$, followed by the sieve, bridge, and envelope rates, gives $\|\boldsymbol{\rho}_{\tilde{k}}\|_\infty = o_p(\sqrt{T})$. Hence

$$\max_t |\widehat{Z}_t(\beta_0, \tilde{\Lambda})| = o_p(\sqrt{T}).$$

The estimated-partition convex-hull condition and Lemma 3 now yield

$$\ell_T(\beta_0, \tilde{\Lambda}) = \frac{A_T(\tilde{\Lambda})^2}{B_T(\tilde{\Lambda})} + o_p(1).$$

Substitution of the two equivalences reduces this ratio to the true-partition ratio plus $o_p(1)$, and Theorem 1 gives the χ_1^2 limit. $\square$

7 Conditional-Design Profile-Sieve Segmentation

This section is the main rebuild of the Bai–Perron module. The break search is no longer imported as a broad high-level assumption. Instead, the stochastic part is controlled by conditional-on-design sub-Gaussian concentration and the identifying part is isolated as a deterministic profile-loss condition.

At the null value β_0 , define

$$y_t(\beta_0) = Y_t - \beta_0 X_{t-1}.$$

Then the nuisance-break model and sieve approximation imply

$$y_t(\beta_0) = \mathbf{P}_K(W_{t-1})' \mathbf{c}_{j,K} + U_t + r_{j,K,t}, \quad k_{j,0} < t \leq k_{j+1,0}.$$

Thus, apart from approximation error, the break search is a segmented least-squares problem with regime-specific sieve coefficient vectors.

Assumption 15 (Conditional-design profile-sieve segmentation). *For the main estimated-partition theory, $q_0 = 1$, $\Lambda_0 = (k_0)$, and $\bar{q}$ is fixed. Let $\boldsymbol{\mu}_K = \mathbf{Q}_{K,\Lambda_0} \mathbf{c}_0$ be the sieve part of the true conditional mean and define*

$$D_T(\Lambda) = \|\mathbf{M}_{K,\Lambda} \boldsymbol{\mu}_K\|^2.$$

The following conditions hold.

1. **Admissible partitions.** All searches are over $\mathcal{L}_q(\epsilon)$, the true break obeys the stricter spacing $\epsilon_0 > \epsilon$, and $(\bar{q} + 1)\epsilon < 1$.
2. **Global segment conditioning.** The global Gram condition in Assumption 5 holds for every admissible segment used in the search.
3. **Deterministic profile identification.** For every admissible one-break candidate,

$$D_T(k) \geq c_\Delta |k - k_0| \Delta_T^2.$$

For the underfitted zero-break model,

$$\inf_{\Lambda \in \mathcal{L}_0(\epsilon)} D_T(\Lambda) \geq c_\Delta T \Delta_T^2.$$

4. **Uniform conditional RSS expansion.** With

$$R_T = K + \log T + Ta_K^2, \quad r_T = R_T/\Delta_T^2,$$

conditional sub-Gaussian concentration gives, uniformly over the admissible one-break search,

$$RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0) \geq \frac{1}{2}D_T(k) - O_p(R_T).$$

It also gives the order-selection comparisons

$$RSS_K\{\beta_0, \hat{\Lambda}_0(\beta_0)\} - RSS_K(\beta_0, k_0) \geq cT\Delta_T^2 - O_p(R_T),$$

and, uniformly over $1 < q \leq \bar{q}$,

$$RSS_K(\beta_0, k_0) - RSS_K\{\beta_0, \hat{\Lambda}_q(\beta_0)\} = O_p(R_T).$$

The rates satisfy

$$R_T = o(T\Delta_T^2), \quad r_T = o(\sqrt{T}).$$

Proposition 1 (Primitive verification of the profile-RSS expansion). *Suppose $q_0 = 1$, $\bar{q}$ is fixed, and Assumptions 1, 5, 6, and 14 hold. Then the uniform RSS expansion and the underfit/overfit order comparisons in Assumption 15 hold with $R_T = K + \log T + Ta_K^2$.*

Detailed solution. Condition on $\mathcal{D}_T$ and work on the Gram and sieve-approximation events whose probability tends to one. For any admissible partition Λ , write $\mathbf{y}(\beta_0) = \boldsymbol{\mu}_K + \mathbf{r}_0 + \mathbf{U}$. Since $\mathbf{M}_{K, \Lambda_0} \boldsymbol{\mu}_K = \mathbf{0}$,

$$RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0) = D_T(\Lambda) + 2\mathbf{U}'\mathbf{M}_{K, \Lambda}\boldsymbol{\mu}_K + \mathbf{U}'(\mathbf{M}_{K, \Lambda} - \mathbf{M}_{K, \Lambda_0})\mathbf{U} + \mathcal{E}_{r, T}(\Lambda),$$

where $\mathcal{E}_{r, T}(\Lambda)$ collects every term containing $\mathbf{r}_0$. The admissible search over $0 \leq q \leq \bar{q}$ has polynomial cardinality, so its logarithm is $O(\log T)$.

For fixed Λ , conditional sub-Gaussian tails give

$$2|\mathbf{U}'\mathbf{M}_{K, \Lambda}\boldsymbol{\mu}_K| \leq \frac{1}{4}D_T(\Lambda) + O_p(\log T)$$

uniformly over the search after a union bound, because the conditional variance proxy is proportional to $D_T(\Lambda)$. Since $\mathbf{M}_{K, \Lambda} - \mathbf{M}_{K, \Lambda_0} = \mathbf{\Pi}_{K, \Lambda_0} - \mathbf{\Pi}_{K, \Lambda}$, it is a difference of two fitted-space projections whose ranks are at most $(\bar{q} + 1)K$. Hence the difference has rank $O(K)$ and operator norm bounded by a constant. Hanson–Wright concentration gives

$$\sup_{\Lambda} |\mathbf{U}'(\mathbf{M}_{K, \Lambda} - \mathbf{M}_{K, \Lambda_0})\mathbf{U} - \mathbb{E}\{\mathbf{U}'(\mathbf{M}_{K, \Lambda} - \mathbf{M}_{K, \Lambda_0})\mathbf{U} \mid \mathcal{D}_T\}| = O_p(K + \log T),$$

and the conditional expectation is $O(K)$ by the bounded conditional variances and the same rank bound.

Finally, projection contraction and $\|\mathbf{r}_0\|^2 = O_p(Ta_K^2)$ imply, uniformly over the search,

$$|\mathcal{E}_{r, T}(\Lambda)| \leq \frac{1}{4}D_T(\Lambda) + O_p(Ta_K^2 + K + \log T).$$

The mixed term $\mathbf{r}'_0(\mathbf{M}_{K,\Lambda} - \mathbf{M}_{K,\Lambda_0})\mathbf{U}$ is absorbed by $2ab \leq a^2 + b^2$ and the preceding projection-noise bound. Combining the three displays yields

$$RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0) \geq \frac{1}{2}D_T(\Lambda) - O_p(R_T)$$

uniformly over the admissible search.

For one-break candidates, the deterministic identification inequality in Assumption 14 gives the displayed one-break RSS expansion and the localization rate $r_T = R_T/\Delta_T^2$. For the zero-break model, the underfit identification bound gives

$$RSS_K\{\beta_0, \hat{\Lambda}_0(\beta_0)\} - RSS_K(\beta_0, k_0) \geq cT\Delta_T^2 - O_p(R_T).$$

For $q > 1$, the same uniform lower bound and nonnegativity of D_T give

$$RSS_K(\beta_0, k_0) - RSS_K\{\beta_0, \hat{\Lambda}_q(\beta_0)\} = O_p(R_T),$$

because the right side is the largest possible stochastic overfit gain relative to the true one-break partition. These are exactly the order-selection comparisons in Assumption 15. $\square$

Proposition 2 (Profile-sieve reduction to segmented least squares). *At β_0 , the fixed-order nuisance-break estimator is the least-squares segmentation estimator in the generated regression*

$$y_t(\beta_0) = \mathbf{P}_K(W_{t-1})'\mathbf{c}_j + \text{error}, \quad t \in I_j(\Lambda).$$

Detailed solution. For a fixed candidate partition Λ , minimizing the null-imposed segment cost over the nuisance coefficients gives

$$RSS_K(\beta_0, \Lambda) = \sum_{j=0}^q \min_{\mathbf{c}_j \in \mathbb{R}^K} \sum_{t=k_j+1}^{k_{j+1}} \{y_t(\beta_0) - \mathbf{P}_K(W_{t-1})'\mathbf{c}_j\}^2.$$

This is the segmented least-squares criterion for a linear regression whose coefficient vector may change across regimes. The block matrix $\mathbf{Q}_{K,\Lambda}$ is the stacked design matrix and $\mathbf{M}_{K,\Lambda}$ is its residual-maker. The nonstandard work is exactly what Assumption 15 records: growing K , approximation error, uniform segment conditioning, and conditional concentration over the finite admissible search. $\square$

Lemma 15 (One-break profile-RSS exclusion). *Under Assumption 15, for every fixed M , define*

$$\mathcal{B}_T(M) = \{k \in \mathcal{L}_1(\epsilon) : |k - k_0| \geq Mr_T\}.$$

Then

$$\lim_{M \rightarrow \infty} \liminf_{T \rightarrow \infty} \mathbb{P} \left\{ \inf_{k \in \mathcal{B}_T(M)} [RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0)] > 0 \right\} = 1.$$

Detailed solution. For $k \in \mathcal{B}_T(M)$, deterministic identification and the uniform RSS expansion give

$$RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0) \geq \frac{1}{2}c_\Delta Mr_T \Delta_T^2 - O_p(R_T) = \left(\frac{1}{2}c_\Delta M + O_p(1) \right) R_T.$$

Taking $T \rightarrow \infty$ and then $M \rightarrow \infty$ proves the exclusion event. $\square$

Theorem 4 (One-break profile-sieve localization). *Under Assumption 15,*

$$|\widehat{k}_1(\beta_0) - k_0| = O_p(r_T), \quad r_T = R_T/\Delta_T^2 = o(\sqrt{T}).$$

Detailed solution. The one-break estimator minimizes $RSS_K(\beta_0, k)$ over $\mathcal{L}_1(\epsilon)$, and k_0 is admissible. On the exclusion event in Lemma 15, no candidate with $|k - k_0| \geq Mr_T$ can minimize the criterion. Hence

$$\mathbb{P}\{|\widehat{k}_1(\beta_0) - k_0| \geq Mr_T\} \leq \mathbb{P}\left\{\inf_{k \in \mathcal{B}_T(M)} [RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0)] \leq 0\right\},$$

and the right side vanishes after taking $T \rightarrow \infty$ and then $M \rightarrow \infty$. $\square$

Assumption 16 (Concrete unknown-order penalty). *Use*

$$\text{pen}_T(q, K) = q\kappa_T R_T,$$

where

$$\kappa_T \rightarrow \infty, \quad \kappa_T R_T = o(T\Delta_T^2).$$

For instance, $\kappa_T = \log \log(T + e^e)$ is valid whenever $R_T \log \log T = o(T\Delta_T^2)$.

Theorem 5 (Unknown-order consistency, one-break truth). *Under Assumptions 15 and 16, with $q_0 = 1$,*

$$\mathbb{P}\{\widehat{q}(\beta_0) = 1\} \rightarrow 1.$$

Detailed solution. Write $RSS_q^* = RSS_K\{\beta_0, \widehat{\Lambda}_q(\beta_0)\}$. For the underfitted order $q = 0$, Assumption 15 gives

$$RSS_0^* - RSS_1^* \geq cT\Delta_T^2 - O_p(R_T).$$

The penalty advantage of underfitting is only $\kappa_T R_T = o(T\Delta_T^2)$, so the penalized zero-break objective is larger with probability tending to one. For every $q > 1$, the overfit RSS gain is $O_p(R_T)$, while the penalty increase is $(q - 1)\kappa_T R_T$. Since $\kappa_T \rightarrow \infty$ and $\bar{q}$ is fixed, all overfitted objectives are larger with probability tending to one. A finite union bound over $q \neq 1$ proves the claim. $\square$

Corollary 1 (Jointly feasible rates). *Suppose $\zeta_K^2 \asymp K$, $a_K \asymp c_K \asymp K^{-s/d}$, $\Delta_T \geq c_\Delta > 0$, and choose*

$$K \asymp T^{d/(2s+d)}.$$

If $s > d$, then

$$R_T = K + \log T + Ta_K^2 \asymp K, \quad r_T = R_T/\Delta_T^2 \asymp K,$$

and all main bridge, local-bias, envelope, convex-hull transfer, and penalty rates hold with, for example,

$$\text{pen}_T(q, K) = qK \log \log(T + e^e).$$

Detailed solution. With $K \asymp T^{d/(2s+d)}$ and $a_K \asymp K^{-s/d}$, $Ta_K^2 \asymp K$, so $R_T \asymp K$ and $r_T \asymp K$. The largest bridge restriction is

$$\frac{K^3 \log K}{T} \rightarrow 0,$$

up to logarithmic factors, and this follows from $s > d$. The local-bias rate is

$$\frac{a_K K^{5/2}}{\sqrt{T}} \rightarrow 0,$$

which is also implied by $s > d$. The score-bias orthogonality route requires

$$\sqrt{T} a_K c_K \rightarrow 0,$$

which already holds under $s > d/2$ and therefore under $s > d$. The convex-hull transfer terms reduce to

$$K^{1-s/d} \rightarrow 0, \quad K \sqrt{\frac{\log T}{T}} \rightarrow 0, \quad \frac{K^{5/2}}{T} \rightarrow 0,$$

again under $s > d$. Finally, $K \log \log T = o(T\Delta_T^2)$, so the displayed penalty satisfies Assumption 16. $\square$

Lemma 16 (High-probability event transfer). *Let X_T and Y_T be real statistics, and let E_T be events such that $\mathbb{P}(E_T) \rightarrow 1$. If $X_T \Rightarrow X$ and $Y_T = X_T$ on E_T , then $Y_T \Rightarrow X$.*

Detailed solution. For every bounded continuous function f ,

$$|\mathbb{E}f(Y_T) - \mathbb{E}f(X_T)| \leq 2\|f\|_\infty \mathbb{P}(E_T^c) \rightarrow 0.$$

Since $\mathbb{E}f(X_T) \rightarrow \mathbb{E}f(X)$, the same convergence holds for $\mathbb{E}f(Y_T)$, which is weak convergence of Y_T to X . $\square$

Corollary 2 (Unknown-order Wilks, one-break truth). *Suppose $q_0 = 1$, Theorem 5 holds, the fixed-order estimated-partition Wilks theorem 3 holds for $\tilde{\Lambda} = \hat{\Lambda}_1(\beta_0)$, and the same measurable tie-breaking convention is used for fixed-order and unknown-order selection. Then*

$$\ell_T\{\beta_0, \hat{\Lambda}(\beta_0)\} \Rightarrow \chi_1^2.$$

Detailed solution. Let $E_T = \{\hat{q}(\beta_0) = 1\}$. By Theorem 5, $\mathbb{P}(E_T) \rightarrow 1$. On E_T , unknown-order selection uses the one-break order and the same deterministic tie-breaking rule as the fixed-order estimator, so

$$\hat{\Lambda}(\beta_0) = \hat{\Lambda}_1(\beta_0) = \tilde{\Lambda}.$$

Therefore

$$\ell_T\{\beta_0, \hat{\Lambda}(\beta_0)\} = \ell_T(\beta_0, \tilde{\Lambda})$$

on E_T . The fixed-order estimated-partition Wilks theorem gives $\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2$, and Lemma 16 transfers this limit to the unknown-order statistic. $\square$

Remark 4 (Proof audit for Corollary 2). The corollary uses only order consistency and event transfer, and it is stated only for the one-break truth covered by Theorem 3. It does not introduce a new empirical-likelihood expansion. The deterministic tie-breaking convention is essential: without it, the selected partition on E_T need not be literally the same measurable object as $\hat{\Lambda}_1(\beta_0)$.

Theorem 6 (Master one-break conditional-design Wilks). *Suppose $q_0 = 1$, $\bar{q}$ is fixed, and Assumptions 1–6, 10, 11, 12, 13, 14, and 16 hold. Suppose also*

$$T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta^2, \quad \mathbb{P}(\eta \geq \sqrt{c_\eta}) = 1$$

for some $c_\eta > 0$, and the bridge rates in Theorem 2 hold. Then, for the fixed one-break profile estimator $\tilde{\Lambda} = \hat{\Lambda}_1(\beta_0)$,

$$\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2.$$

Moreover, for the unknown-order selector over $0 \leq q \leq \bar{q}$ with the same deterministic tie-breaking rule,

$$\ell_T\{\beta_0, \hat{\Lambda}(\beta_0)\} \Rightarrow \chi_1^2.$$

Detailed solution. Proposition 1 verifies Assumption 15. Theorem 4 then verifies the localization input in Assumption 8 for $\tilde{\Lambda} = \hat{\Lambda}_1(\beta_0)$. Theorem 2 then transfers the first two empirical-score moments from Λ_0 to $\tilde{\Lambda}$. Lemma 2 verifies the convex-hull condition for both the true and estimated one-break scores under Assumption 10 and the displayed transfer rate. The known-partition Wilks theorem and the estimated-partition bridge therefore give $\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2$.

For unknown order, Theorem 5 gives $\mathbb{P}\{\hat{q}(\beta_0) = 1\} \rightarrow 1$. On that event the unknown-order partition equals $\tilde{\Lambda}$ by the common tie-breaking convention, so Lemma 16 transfers the fixed-order Wilks limit to the unknown-order statistic. $\square$

8 Local Alternatives

Assumption 17 (Canonical local alternative and drift). *Let*

$$\beta_T = \beta_0 + \frac{h}{\sqrt{T}}, \quad h \in \mathbb{R}$$

with deterministic h . Define

$$\Gamma_T = T^{-1} \mathbf{X}'_{lag} \mathbf{M}_{K, \Lambda_0} \mathbf{w} = T^{-1} \sum_{t=1}^T (\mathbf{M}_{K, \Lambda_0} \mathbf{X}_{lag})_t v_{t,0}.$$

Under P_{β_T} , Assumption 1 and the Gram/envelope conditions continue to hold, and

$$\Gamma_T \xrightarrow{p} \Gamma, \quad \eta_T^2 := T^{-1} \sum_{t=1}^T \sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta^2, \quad 0 < \eta^2 < \infty,$$

where Γ and η are deterministic. Also

$$T^{-1} \sum_{t=1}^T \{(\mathbf{M}_{K,\Lambda_0} \mathbf{X}_{lag})_t v_{t,0}\}^2 = O_p(1), \quad \max_t |(\mathbf{M}_{K,\Lambda_0} \mathbf{X}_{lag})_t v_{t,0}| = o_p(T).$$

Put

$$g_{t,T} = \frac{h}{\sqrt{T}} (\mathbf{M}_{K,\Lambda_0} \mathbf{X}_{lag})_t v_{t,0}, \quad \delta_h = \frac{h\Gamma}{\eta}.$$

Assumption 18 (Local drift stability under break perturbation). *For every fixed $C < \infty$, uniformly over $|k - k_0| \leq Cr_T$,*

$$T^{-1/2} |\mathbf{X}'_{lag} \{\mathbf{M}_{K,k} - \mathbf{M}_{K,k_0}\} \mathbf{w}| = o_p(1),$$

where $q_0 = 1$. With

$$g_t(k) = \frac{h}{\sqrt{T}} (\mathbf{M}_{K,k} \mathbf{X}_{lag})_t (\mathbf{M}_{K,k} \mathbf{w})_t,$$

we also have

$$\sup_{|k-k_0| \leq Cr_T} T^{-1} \sum_{t=1}^T \{g_t(k) - g_t(k_0)\}^2 = o_p(1), \quad \sup_{|k-k_0| \leq Cr_T} \max_t |g_t(k) - g_t(k_0)| = o_p(\sqrt{T}).$$

Finally, under P_{β_T} , the profile-RSS expansion in Assumption 15 remains valid with the same R_T .

Proposition 3 (Local-law bridge and order stability). *Suppose $q_0 = 1$, Assumptions 17 and 18 hold, and the null-law one-break localization, bridge, and unknown-order penalty rates hold. Under P_{β_T} ,*

$$|\tilde{k} - k_0| = O_p(r_T),$$

$$S_T(\tilde{\Lambda}) - S_T(\Lambda_0) = o_p(1), \quad V_T(\tilde{\Lambda}) - V_T(\Lambda_0) = o_p(1), \quad \max_t |\hat{Z}_t(\beta_0, \tilde{\Lambda})| = o_p(\sqrt{T}),$$

and, if unknown order is used,

$$P_{\beta_T} \{\hat{q}(\beta_0) = 1\} \rightarrow 1.$$

Detailed solution. Under $\beta_T = \beta_0 + h/\sqrt{T}$, the null-imposed segmentation criterion adds only the local slope-drift terms generated by $h\mathbf{X}_{lag}/\sqrt{T}$. Assumption 18 states precisely that these terms are $o_p(1)$ at the empirical-likelihood score scale and are absorbed into the same $O_p(R_T)$ RSS remainder used under the null. Hence the proof of Theorem 4 and Theorem 5 applies under P_{β_T} . The numerator and denominator bridge proofs gain the drift array $g_t(k) - g_t(k_0)$, and the two displayed drift-stability bounds make its contribution negligible in the first two score moments and in the maximum-score condition. The remaining terms are exactly the null-law bridge terms already controlled in Theorem 2. $\square$

Lemma 17 (Local shifted-score replacement). *Suppose Assumptions 1, 6, 3, 5, and 11 continue to hold under P_{β_T} . Then*

$$T^{-1/2} \sum_{t=1}^T \{\widehat{Z}_t(\beta_0, \Lambda_0) - (U_t v_{t,0} + g_{t,T})\} = o_p(1), \quad (8.1)$$

$$T^{-1} \sum_{t=1}^T \{\widehat{Z}_t(\beta_0, \Lambda_0)^2 - (U_t v_{t,0} + g_{t,T})^2\} = o_p(1), \quad (8.2)$$

$$\max_t |\widehat{Z}_t(\beta_0, \Lambda_0) - (U_t v_{t,0} + g_{t,T})| = o_p(\sqrt{T}). \quad (8.3)$$

Detailed solution. Under P_{β_T} ,

$$\mathbf{Y} - \beta_0 \mathbf{X}_{lag} = \mathbf{Q}_{K, \Lambda_0} \mathbf{c}_0 + \mathbf{r}_0 + \mathbf{U} + \frac{h}{\sqrt{T}} \mathbf{X}_{lag}.$$

Apply $\mathbf{M}_{K, \Lambda_0}$ and put

$$\mathbf{e}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{r}_0, \quad \mathbf{h}_0 = \mathbf{\Pi}_{K, \Lambda_0} \mathbf{U}, \quad \mathbf{x}_0^c = \mathbf{M}_{K, \Lambda_0} \mathbf{X}_{lag}.$$

Then

$$\widehat{\mathbf{u}}(\beta_0, \Lambda_0) = \mathbf{U} - \mathbf{h}_0 + \mathbf{e}_0 + \frac{h}{\sqrt{T}} \mathbf{x}_0^c.$$

Multiplication by $\mathbf{v}_0 = \mathbf{M}_{K, \Lambda_0} \mathbf{w}$ gives the exact identity

$$\widehat{Z}_t(\beta_0, \Lambda_0) = U_t v_{t,0} + g_{t,T} + e_{t,T}^{\text{feas}}, \quad e_{t,T}^{\text{feas}} = (e_{t,0} - h_{t,0}) v_{t,0}.$$

The vector $\mathbf{e}_T^{\text{feas}}$ is the same feasibility error controlled in Lemma 9; under the stated local-law continuation of the primitive assumptions,

$$T^{-1/2} \sum_t e_{t,T}^{\text{feas}} = o_p(1), \quad \|\mathbf{e}_T^{\text{feas}}\|_T = o_p(1), \quad \|\mathbf{e}_T^{\text{feas}}\|_\infty = o_p(\sqrt{T}).$$

The first and third displays of the lemma follow immediately.

For the square replacement, set $a_{t,T} = U_t v_{t,0} + g_{t,T}$. Then

$$T^{-1} \sum_t \{\widehat{Z}_t^2 - a_{t,T}^2\} = 2T^{-1} \sum_t a_{t,T} e_{t,T}^{\text{feas}} + T^{-1} \sum_t (e_{t,T}^{\text{feas}})^2.$$

Assumption 17 gives $\|\mathbf{U} \odot \mathbf{v}_0\|_T = O_p(1)$ and $\|\mathbf{g}_T\|_T = o_p(1)$, hence $\|\mathbf{a}_T\|_T = O_p(1)$. By Cauchy–Schwarz and $\|\mathbf{e}_T^{\text{feas}}\|_T = o_p(1)$, both terms on the right are $o_p(1)$. $\square$

Lemma 18 (True-partition local empirical-likelihood limit). *Under Assumption 17, Lemma 17, and the local convex-hull condition for $\widehat{Z}_t(\beta_0, \Lambda_0)$,*

$$\ell_T(\beta_0, \Lambda_0) \Rightarrow \chi_1^2(\delta_h^2).$$

Detailed solution. Lemma 17, the conditional-design CLT from Lemma 6 under the local law, and Assumption 17 give

$$\begin{aligned} T^{-1/2} \sum_t \widehat{Z}_t(\beta_0, \Lambda_0) &= T^{-1/2} \sum_t U_t v_{t,0} + T^{-1/2} \sum_t g_{t,T} + o_p(1) \\ &\xrightarrow{\text{st}} \eta Z + h\Gamma = \eta(Z + \delta_h). \end{aligned}$$

For the second moment,

$$T^{-1} \sum_t (U_t v_{t,0} + g_{t,T})^2 = T^{-1} \sum_t U_t^2 v_{t,0}^2 + 2T^{-1} \sum_t U_t v_{t,0} g_{t,T} + T^{-1} \sum_t g_{t,T}^2.$$

The first term converges to η^2 and the last term is $o_p(1)$. The cross term is $o_p(1)$ by Cauchy-Schwarz:

$$\left| T^{-1} \sum_t U_t v_{t,0} g_{t,T} \right| \leq \left(T^{-1} \sum_t U_t^2 v_{t,0}^2 \right)^{1/2} \left(T^{-1} \sum_t g_{t,T}^2 \right)^{1/2} = o_p(1).$$

The local replacement lemma therefore gives

$$T^{-1} \sum_t \widehat{Z}_t(\beta_0, \Lambda_0)^2 \xrightarrow{p} \eta^2.$$

The maximum condition follows from

$$\max_t |U_t v_{t,0} + g_{t,T}| = o_p(\sqrt{T})$$

and the maximum replacement bound in Lemma 17. The scalar empirical-likelihood expansion, Lemma 3, now yields

$$\ell_T(\beta_0, \Lambda_0) = \frac{\{\eta(Z + \delta_h)\}^2}{\eta^2} + o_p(1) \Rightarrow (Z + \delta_h)^2 \sim \chi_1^2(\delta_h^2).$$

□

Theorem 7 (Local noncentral chi-square). *Suppose the conditions of Lemma 18 hold, Proposition 3 holds, and the local convex-hull condition holds for $\widehat{Z}_t(\beta_0, \tilde{\Lambda})$. Then*

$$\ell_T(\beta_0, \tilde{\Lambda}) \Rightarrow \chi_1^2(\delta_h^2).$$

If $q_0 = 1$ and $P_{\beta_T}\{\widehat{q}(\beta_0) = 1\} \rightarrow 1$, the same limit holds for $\ell_T\{\beta_0, \widehat{\Lambda}(\beta_0)\}$.

Detailed solution. The proof of Lemma 18 gives the true-partition score limits

$$S_T(\Lambda_0) \xrightarrow{\text{st}} \eta(Z + \delta_h), \quad V_T(\Lambda_0) \xrightarrow{p} \eta^2, \quad \max_t |\widehat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}).$$

The local bridge transfers the first two limits and supplies the maximum bound for the estimated partition:

$$S_T(\tilde{\Lambda}) \xrightarrow{\text{st}} \eta(Z + \delta_h), \quad V_T(\tilde{\Lambda}) \xrightarrow{p} \eta^2, \quad \max_t |\widehat{Z}_t(\beta_0, \tilde{\Lambda})| = o_p(\sqrt{T}).$$

The local convex-hull condition and Lemma 3 give

$$\ell_T(\beta_0, \tilde{\Lambda}) = \frac{S_T(\tilde{\Lambda})^2}{V_T(\tilde{\Lambda})} + o_p(1) \Rightarrow (Z + \delta_h)^2 \sim \chi_1^2(\delta_h^2).$$

For unknown order, let $E_T = \{\hat{q}(\beta_0) = 1\}$. On E_T , $\hat{\Lambda}(\beta_0) = \tilde{\Lambda}$ under the same tie-breaking convention used in Corollary 2. Since $P_{\beta_T}(E_T) \rightarrow 1$, the event-transfer lemma 16 transfers the fixed-order local limit to the unknown-order statistic. $\square$

Remark 5 (Proof audit for Theorem 7). The local theorem has two separate pieces. Lemma 18 proves the noncentral limit at the true partition from the calibrated local drift $g_{t,T}$, the local denominator stability, and the scalar EL expansion. Proposition 3 then moves the result from Λ_0 to $\tilde{\Lambda}$ and supplies local order consistency for unknown-order transfer.

9 Multiple-Break Extension Sketch

The main estimated-partition Wilks theorem is deliberately one-break. For a fixed number $q_0 > 1$, the natural extension is a telescoping argument rather than a new empirical-likelihood expansion. For a local candidate $\Lambda = (k_1, \dots, k_{q_0})$, define

$$\Lambda^{(0)} = \Lambda_0,$$

and

$$\Lambda^{(j)} = (k_1, \dots, k_j, k_{j+1,0}, \dots, k_{q_0,0}), \quad j = 1, \dots, q_0.$$

Then

$$\mathbf{M}_{K,\Lambda} - \mathbf{M}_{K,\Lambda_0} = \sum_{j=1}^{q_0} \{\mathbf{M}_{K,\Lambda^{(j)}} - \mathbf{M}_{K,\Lambda^{(j-1)}}\}.$$

Each summand moves one boundary between two long adjacent segments. If the true breaks are separated and q_0 is fixed, the one-break boundary, coefficient-update, local-bias, and projection-noise arguments can be applied to each summand, with only fixed multiplicative constants. The local candidate set has cardinality $O(r_T^{q_0})$, so the uniform projection bounds replace $\log(1 + r_T)$ by $q_0 \log(1 + r_T) = O(\log r_T)$.

This sketch is not used in the main theorem. A paper that claims general fixed- q_0 estimated-partition Wilks should extend the proof in the following order.

1. **Known multiple-break Wilks.** First keep Λ_0 known and fixed with fixed q_0 . The existing block-projection algebra already covers this case once the Gram, variance, convex-hull, and sieve conditions are stated uniformly over the known blocks.
2. **Fixed-order estimated bridge.** Next hold q_0 fixed and prove the componentwise bridge

$$\max_{1 \leq j \leq q_0} |\hat{k}_j - k_{j,0}| = O_p(r_T), \quad S_T(\hat{\Lambda}) - S_T(\Lambda_0) = o_p(1), \quad V_T(\hat{\Lambda}) - V_T(\Lambda_0) = o_p(1).$$

This is where the telescoping display above must become a lemma, including the maximum-score condition for the scalar empirical-likelihood expansion.

3. **Multi-boundary localization.** Prove localization under fixed q_0 , fixed $\bar{q}$, minimum spacing, minimum jump size Δ_T , and non-overlapping Cr_T neighborhoods around the true breaks. The search-uniform stochastic terms should still be of order $R_T = K + \log T + Ta_K^2$ because q_0 is fixed.
4. **Unknown-order consistency.** Use the same penalty

$$\text{pen}_T(q, K) = q\kappa_T R_T.$$

Underfitted models must lose order $T\Delta_T^2$, while overfitted models may gain only $O_p(R_T)$. This is the multi-break analogue of Theorem 5.

5. **Final multi-break Wilks theorem.** Only after the fixed-order bridge and order-consistency proof are complete should the paper state

$$\ell_T\{\beta_0, \hat{\Lambda}(\beta_0)\} \Rightarrow \chi_1^2.$$

Until those steps are written as full proofs, multi-break computations are exploratory diagnostics rather than main-theorem evidence.

10 End-to-End Assumption Audit

This final audit records the rebuilt paper architecture. The main theorem is not a general martingale, general-multiple-break result. It is a conditional- on-design, sub-Gaussian, one-break estimated-partition Wilks theorem, with unknown-order selection allowed over $q = 0, 1, \dots, \bar{q}$ when the true order is $q_0 = 1$.

Proof block	Required inputs	What it delivers
Conditional-design environment	Conditional independence, mean zero, bounded variance, and sub-Gaussian tails given $\mathcal{D}_T$	Full-sample residualized scores and selected partitions are design-measurable, avoiding predictable-score assumptions
Known-partition Wilks	Variance convergence, nondegeneracy, score envelope, convex hull, and sieve feasibility	$\ell_T(\beta_0, \Lambda_0) \Rightarrow \chi_1^2$ for fixed known q_0
One-break segmentation	Global Gram stability, deterministic profile identification, $R_T = K + \log T + Ta_K^2$, and $r_T = R_T/\Delta_T^2 = o(\sqrt{T})$	$ \hat{k} - k_0 = O_p(r_T)$ under $q_0 = 1$
Estimated-break bridge	Boundary size, local bias stability, local full rank, search-uniform projection noise, and bridge rates	$S_T(\hat{k}) - S_T(k_0) = o_p(1)$ and $V_T(\hat{k}) - V_T(k_0) = o_p(1)$
Unknown order	$\text{pen}_T(q, K) = q\kappa_T R_T$, $\kappa_T \rightarrow \infty$, and $\kappa_T R_T = o(T\Delta_T^2)$	$\hat{q}(\beta_0) = 1$ with probability tending to one
Local alternatives	$\beta_T = \beta_0 + h/\sqrt{T}$, $\Gamma_T \rightarrow_p \Gamma$, $\eta_T^2 \rightarrow_p \eta^2$, and the local bridge	Noncentral limit $\chi_1^2(h^2\Gamma^2/\eta^2)$
Multiple breaks	Telescoping over boundaries plus uniform multi-boundary localization	Extension sketch only; not part of the main theorem

Audit conclusion. The rebuilt manuscript now has one probabilistic spine. Conditional on $\mathcal{D}_T$, the EL score weights and the break search are fixed-design objects, so the stochastic tools are conditional CLT, conditional LLN, and finite-search sub-Gaussian concentration. The price is explicit: the main estimated-break Wilks and unknown-order Wilks results are stated for one nuisance break. A general fixed- q_0 claim requires the telescoping extension in Section 9 to be written as a full proof.

11 Closure Checklist After Editor–Author Loop

The harsh-editor objections addressed in this rebuild are recorded as closure checks rather than future work.

Editor objection	Author repair	Location
The stochastic environment mixed martingale scores with full-design break search	Replaced it by conditional independence and sub-Gaussian tails given $\mathcal{D}_T$	Assumption 1
The residualized instrument was not predictable	Made all projected scores design-measurable conditional-on- $\mathcal{D}_T$ objects	Assumption 2
The order theorem needed $R_T = o(T\Delta_T^2)$ and a concrete penalty	Set $R_T = K + \log T + Ta_K^2$, $r_T = R_T/\Delta_T^2$, and $\text{pen}_T(q, K) = q\kappa_T R_T$	Assumptions 15 and 16
The underfit RSS gap was implicit	Added an explicit zero-break comparison with $cT\Delta_T^2 - O_p(R_T)$	Assumption 15
The random-index projection bounds were pointwise	Added search-uniform T -norm and sup-norm projection-noise bounds	Lemma 11
The segmentation module still looked asserted rather than verified	Added a conditional-concentration proof of the uniform RSS expansion and order comparisons	Proposition 1
The score-bias condition was still imposed without a primitive route	Added a segmentwise orthogonality and smooth residualized-instrument verification	Assumption 7 and Lemma 1
Convex hull was a black-box EL condition	Added a primitive sign-mass condition and convex-hull verification lemma	Assumption 10 and Lemma 2
The endpoint theorem was implicit across several modules	Added a master one-break conditional-design Wilks theorem	Theorem 6
The rates might be mutually empty	Added a concrete smoothness/dimension regime with $s > d$	Corollary 1
Local power reused null-law bridges without saying why	Added local drift stability and local-law bridge/order transfer	Assumption 18 and Proposition 3
The estimated-partition theorem overclaimed multiple breaks	Restricted the main result to $q_0 = 1$ and moved fixed- q_0 to a telescoping sketch	Section 9

Final editorial status. Within the declared scope—conditional-on-design errors, one unknown nuisance break for estimated-partition Wilks, fixed $\bar{q}$, and the displayed rate regime—the workbook now states a coherent proof chain to known-break Wilks, estimated-break Wilks, and unknown-order transfer. The local noncentral result is also stated, but explicitly under the additional local drift-stability and local convex-hull inputs. The only material not claimed as a theorem is the general fixed-multiple-break extension, which is explicitly labeled as an extension sketch rather than part of the main result.